Osteopathic Considerations in Pulmonary Disease – 1

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Make.
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Innovate.
Reinvent the Future.

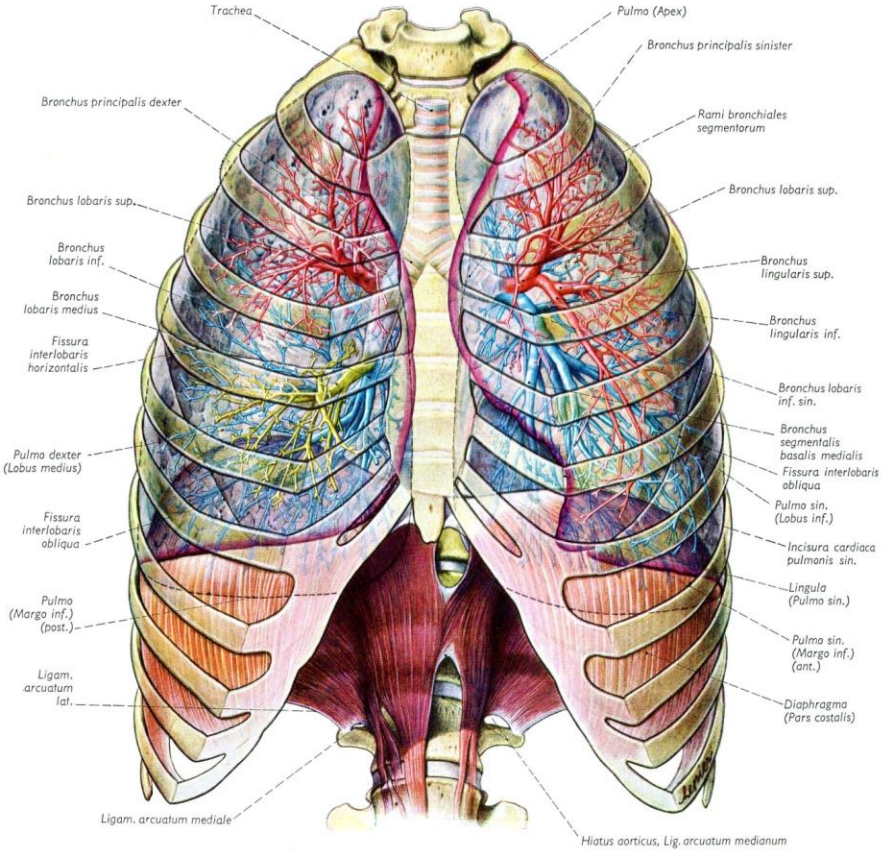
Session Objectives

1) Osteopathically relate physics to pulmonary function and the respiratory-circulatory system - Understand concepts such as: compliance, work of breathing, airway or thoracic cage resistance, respiratory muscle fatigue, Starling's Principle, anaerobic metabolism and paradoxical motion of the ribs.
2) Understand how somatic dysfunction alters pump and bucket handle mechanics, and thereby affects work of breathing.
3) Describe the importance of compliance of the ribcage as a mechanical function that can be evaluated and affected osteopathically.
4) Understand how to assess a patient's breathing based on these concepts

The Science of Osteopathy

- The application of the laws of physics to the physiologic functioning of the body.
- Mechanical principles apply to more than the musculoskeletal system.
- The Pulmonary system is an example of a viscera governed more by mechanics than chemistry.

Forces producing air exchange



Source: Pernkopf atlas of anatomy

Red = sup. lobar bronchus, right and left
Yellow = middle lobar bronchus
Blue = inf. lobar bronchus, right and left

- The negative intrathoracic pressure which produces air exchange is created by the action of skeletal muscles on the osseous thorax. The lung parenchyma is passive

The work of breathing

- The work of breathing is the amount of energy required to produce air exchange.
- It is the sum of airway resistance, parenchymal resistance, and tissue resistance.
- Airway resistance is increased in obstructive diseases, parenchymal resistance is increased in restrictive diseases, tissue resistance is increased in both.

Compliance

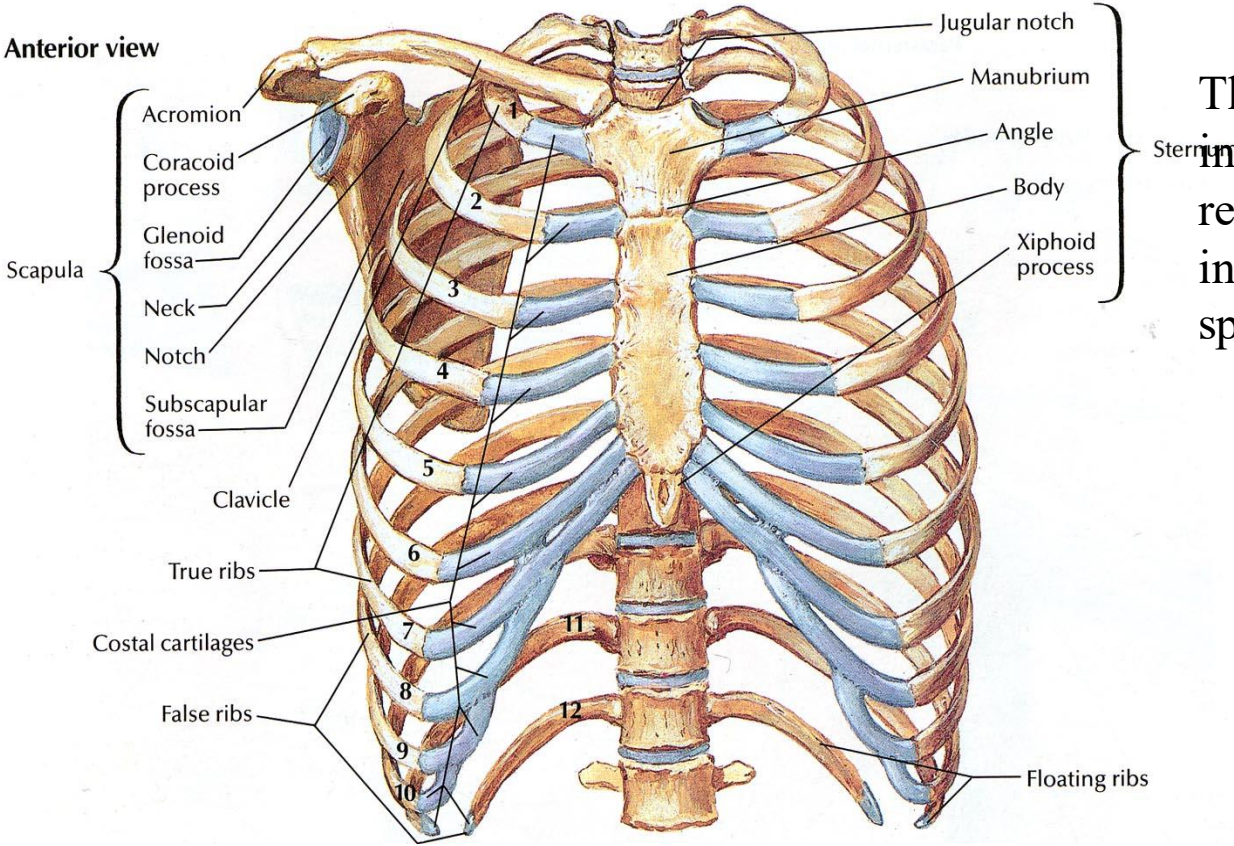
- The term compliance is a term used to describe changes in resistance.
- Increases in airway and parenchyma resistance REDUCE lung compliance.
- Somatic dysfunction and myofascial tension on the thorax and spine REDUCE thoracic cage compliance.

Compliance

- Medical treatment will improve lung compliance. OMT may also be used to improve lung compliance.
- Only OMT will affect thoracic cage compliance. In fact, only osteopathic physicians are trained to evaluate compliance changes in the thoracic cage.

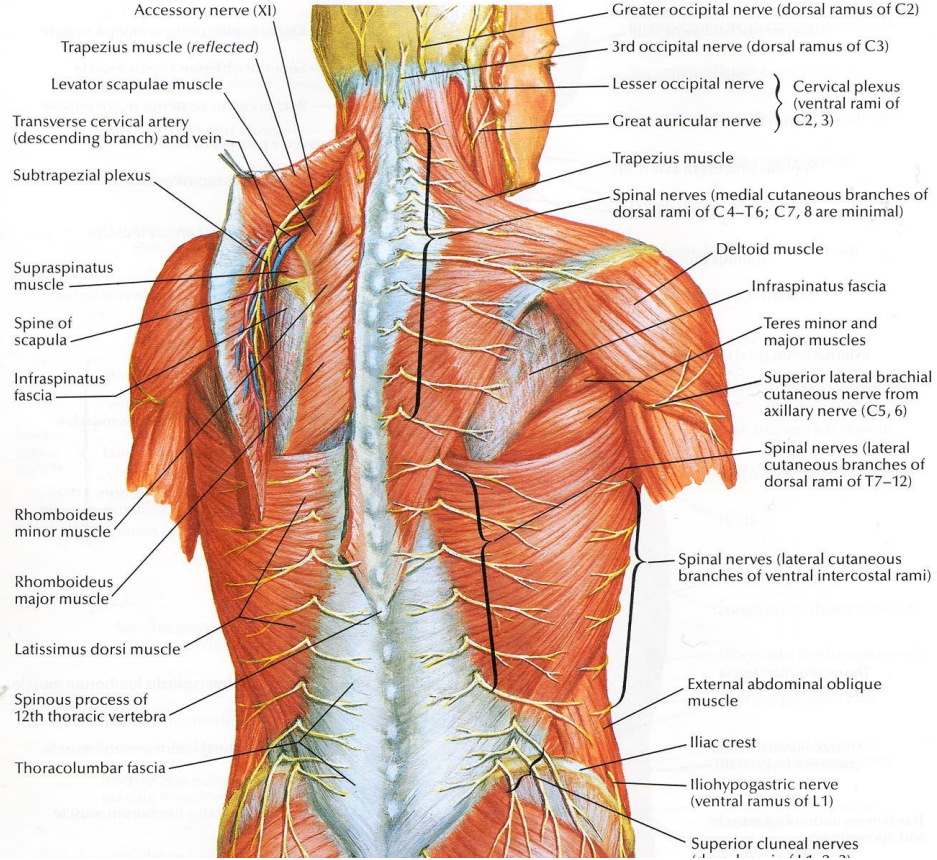
The work of breathing

- In normal breathing, the work of breathing takes approximately 2% of our bodies metabolic energy.
- In severe obstructive or restrictive disease, this number can increase by 15 times, to a full one third of our bodies total energy.
- This produces the severe activity restrictions of pulmonary patients, and makes the work of breathing clinically important in pulmonary disease.



Thoracic cage compliance includes movement and resistance at the joints, including the ribs, the entire spine and pelvis.

Source: Netter atlas of Human Anatomy



Thoracic cage compliance also includes the tension in all the myofascial structures attaching the spine and rib cage.

Source: Netter atlas of Human Anatomy

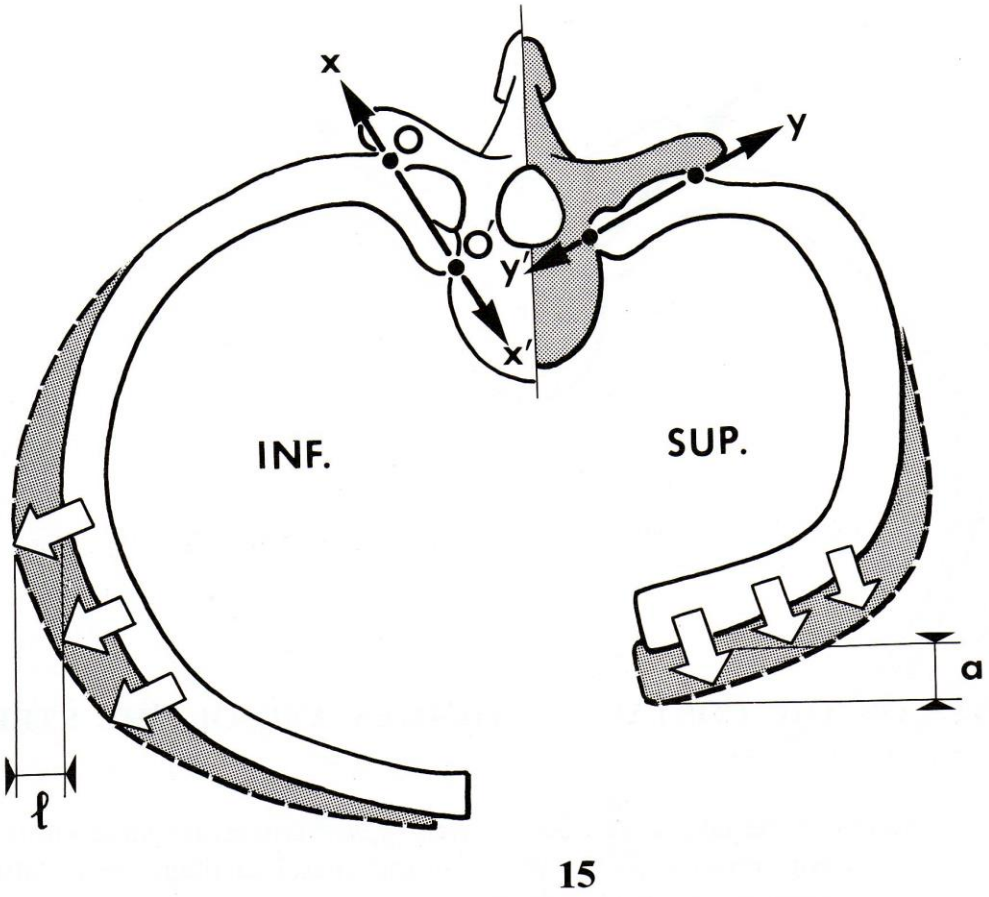
SKELETAL CONSIDERATIONS

Somatic Dysfunction

- By its definition, somatic dysfunction will reduce the compliance of the thorax, and therefore increase the work of breathing.
- In addition, the general shape changes which accompany pulmonary disease, such as the barrel shape in COPD, can create severe decreases in chest wall compliance. These are best addressed with general techniques, as will be shown in lab.

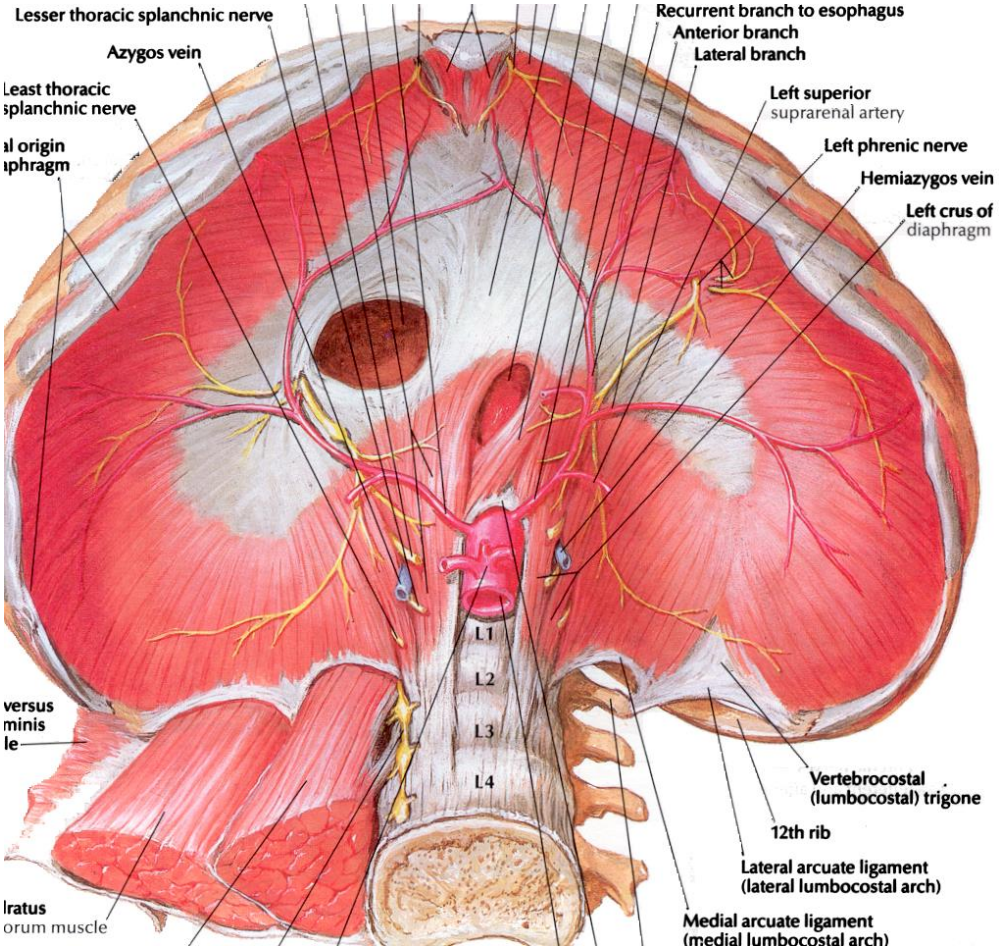
Somatic Dysfunction

- Specific somatic dysfunctions may have an effect on the function of the respiratory mechanism that go beyond the small changes in compliance they invariably produce.



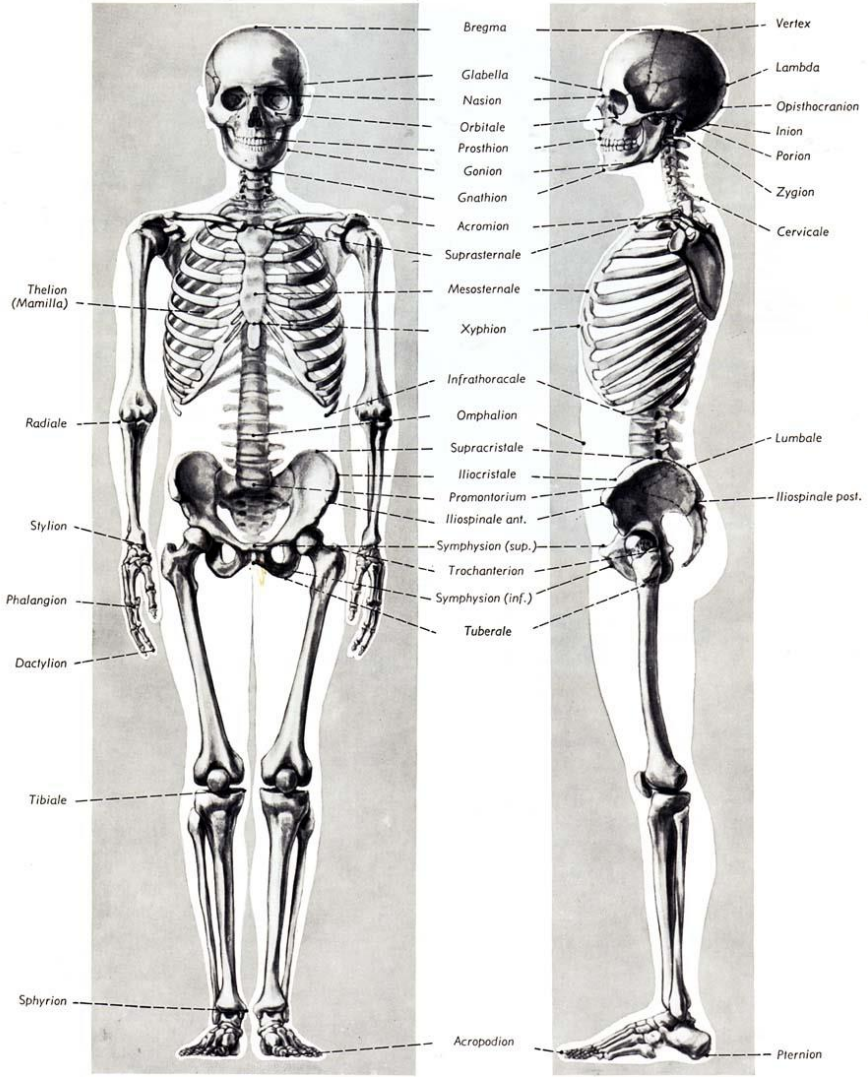
The orientation of the vertebrae and rib help determine the movement allowed at the rib. Vertebral rotation can alter these relations, and produce dyscoordinate breathing patterns.

Souce: Kapandji Vol 3

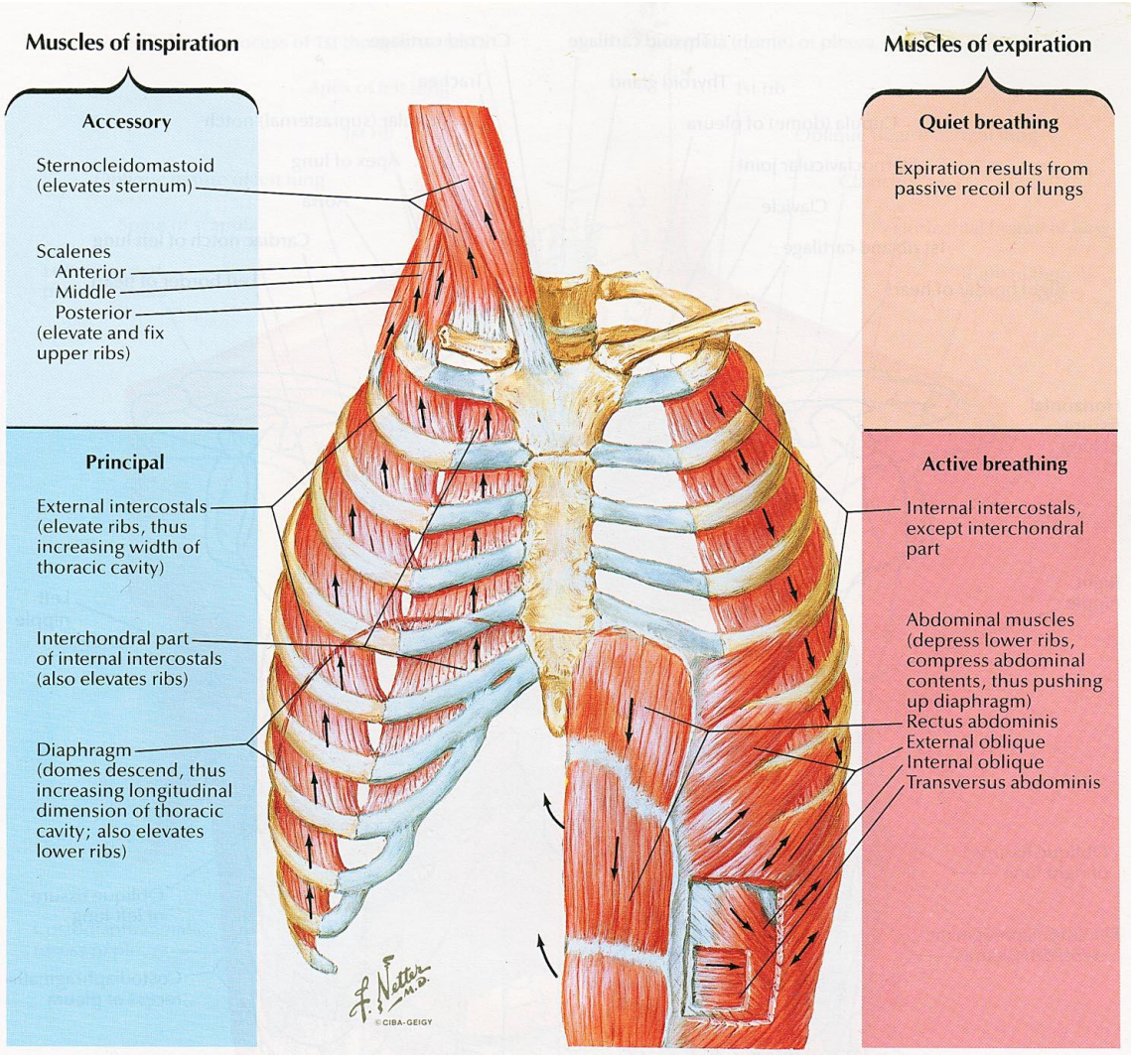


Source: Netter atlas of Human Anatomy

Rotation of the lumbar spine can alter the orientation of the crura, and therefore the function of the diaphragm.



The entire spine, rib cage, and pelvis move with breathing. Dysfunction in any area will have some impact on respiration.



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The Respiratory Musculature

- Respiratory muscle fatigue has been implicated in the pathophysiology of respiratory failure.
- Muscular fatigue is the immediate factor leading to the demise of patients with acute asthma.
- Most signs of respiratory failure are actually signs of respiratory muscle fatigue.

Respiratory muscle fatigue

- Respiratory muscle fatigue has also been identified in COPD, leading to the identification of chronic respiratory muscle fatigue.
- Respiratory muscle fatigue has been implicated in pulmonary edema, lung shock, and difficulty weaning patients off ventilators.

Treating respiratory muscle fatigue

- Treatment to improve the compliance of the thorax will reduce the workload on the muscles, and therefore reduce the severity of fatigue.
- OMT may also be done to directly improve the function and efficiency of the respiratory musculature.

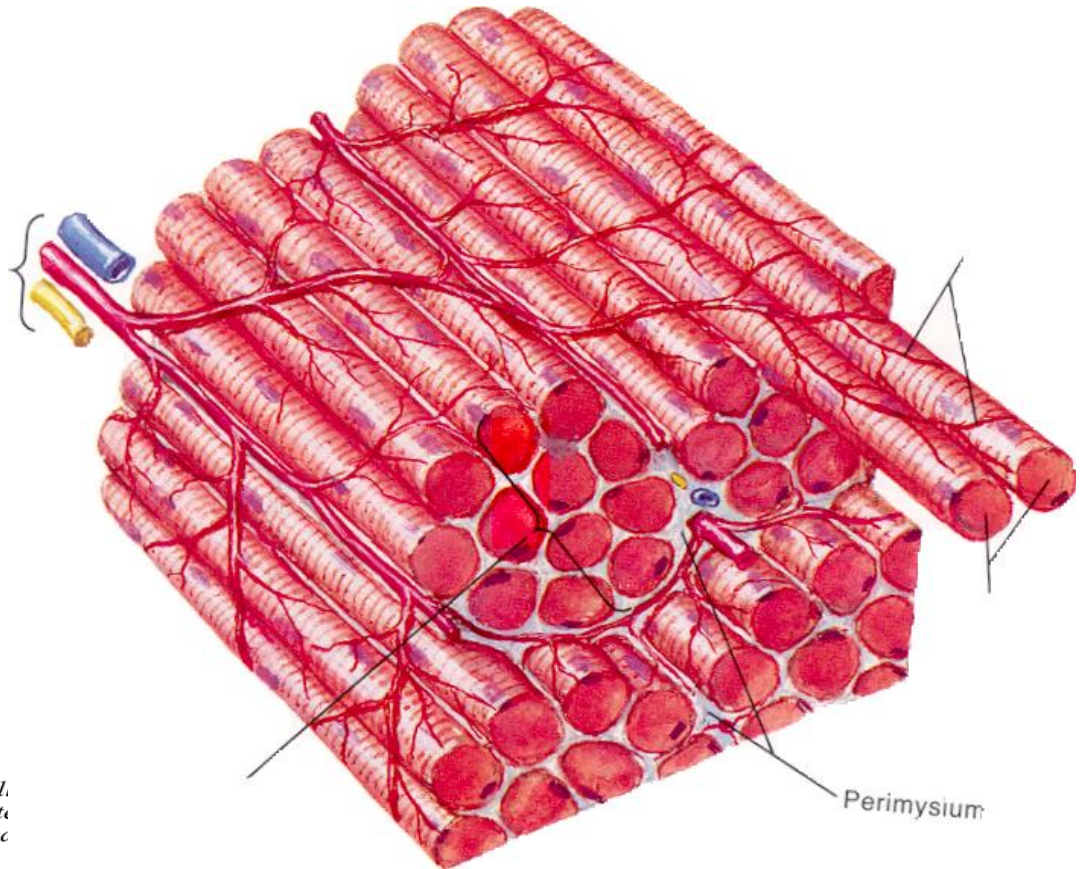
Length-Tension Relationship

- All skeletal muscles operate by Starling's law, the longer the resting length, the stronger the contraction.
- Typical findings in obstructive lung disease produce inefficient muscle contractions
    - -- flattened diaphragm
    - -- short, hypertrophied accessory muscles

Length-Tension Relationship

- The barrel chest of hyperinflation will prevent muscles from returning to normal resting length during exhalation.
- Shortened respiratory muscles will help produce and maintain the barrel shape.
- Increased resting tone also interferes with the blood supply to the muscle.

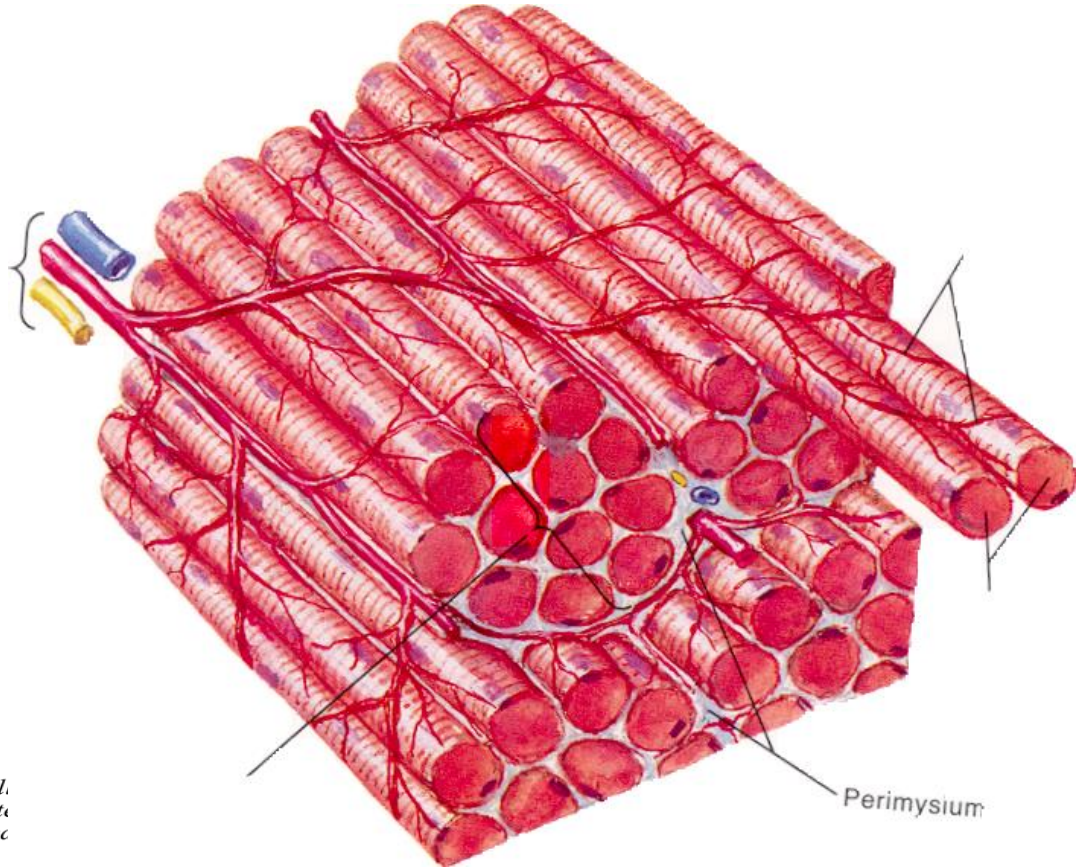
Blood Supply



- Increased oxygen demand with decreased supply will lead to anaerobic metabolism.

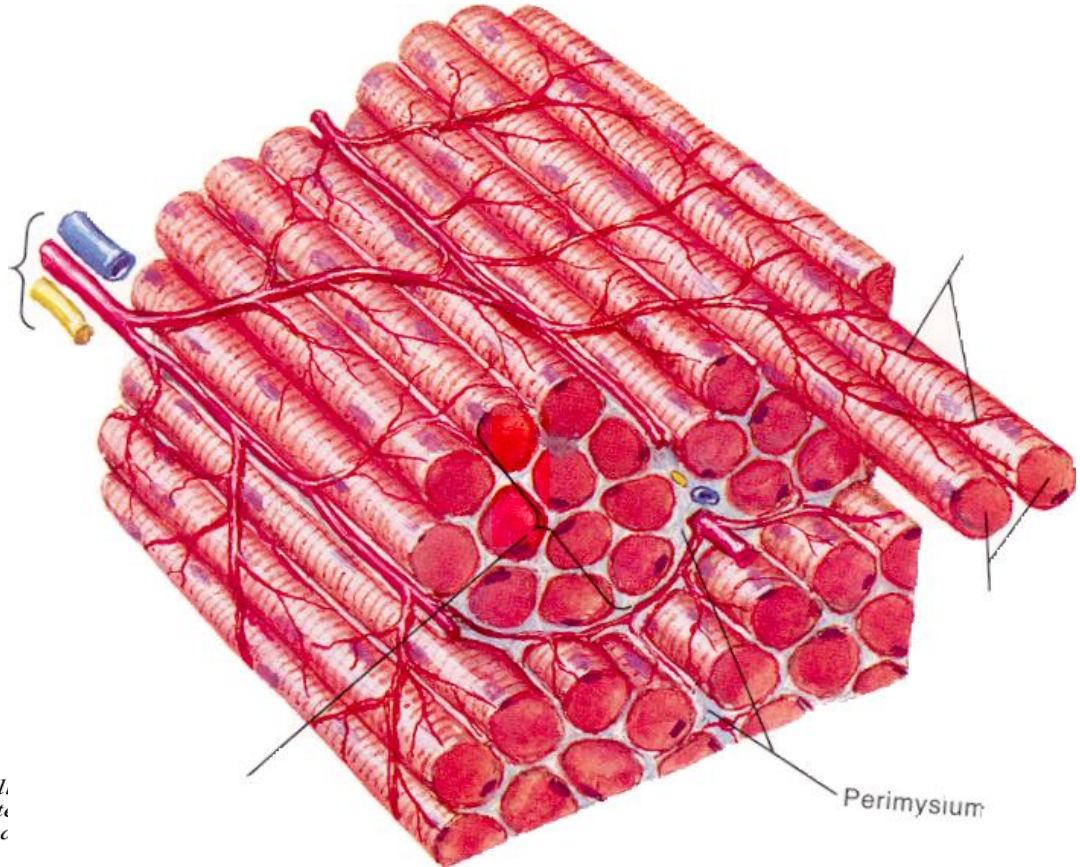
This has been demonstrated in fatigued diaphragms of COPD patients.

Blood Supply



•Anaerobic metabolism is as much as 15 times less efficient than aerobic metabolism.

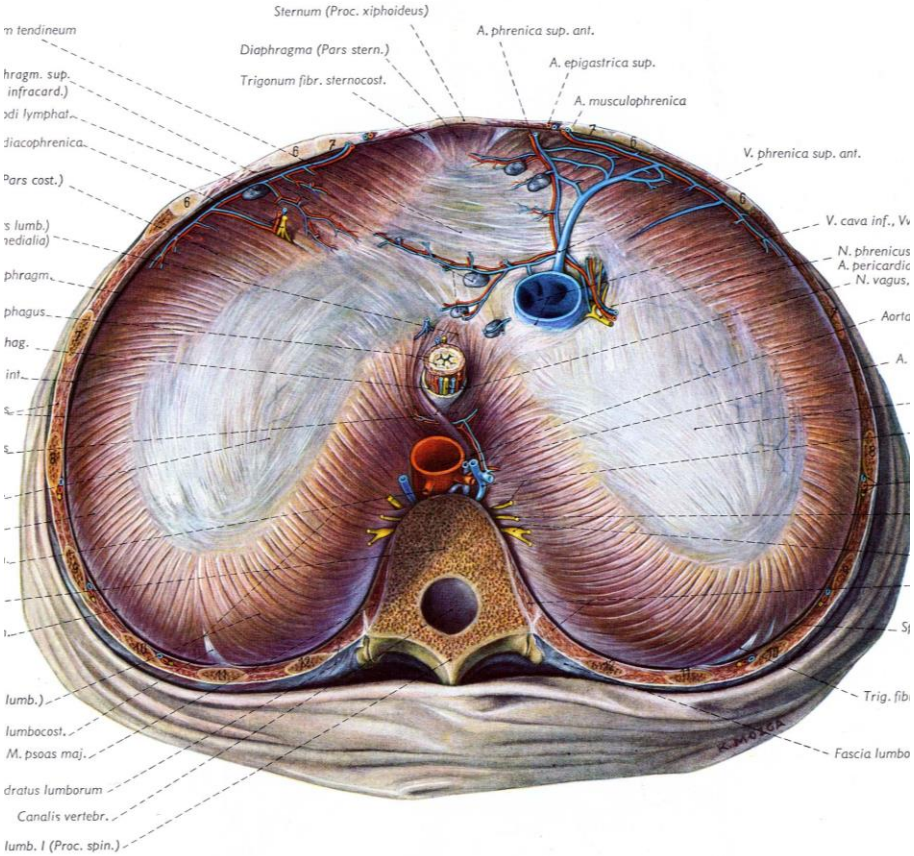
Blood Supply



•OMT to increase length and decrease resting tone is important in the treatment of acute and chronic lung disease.

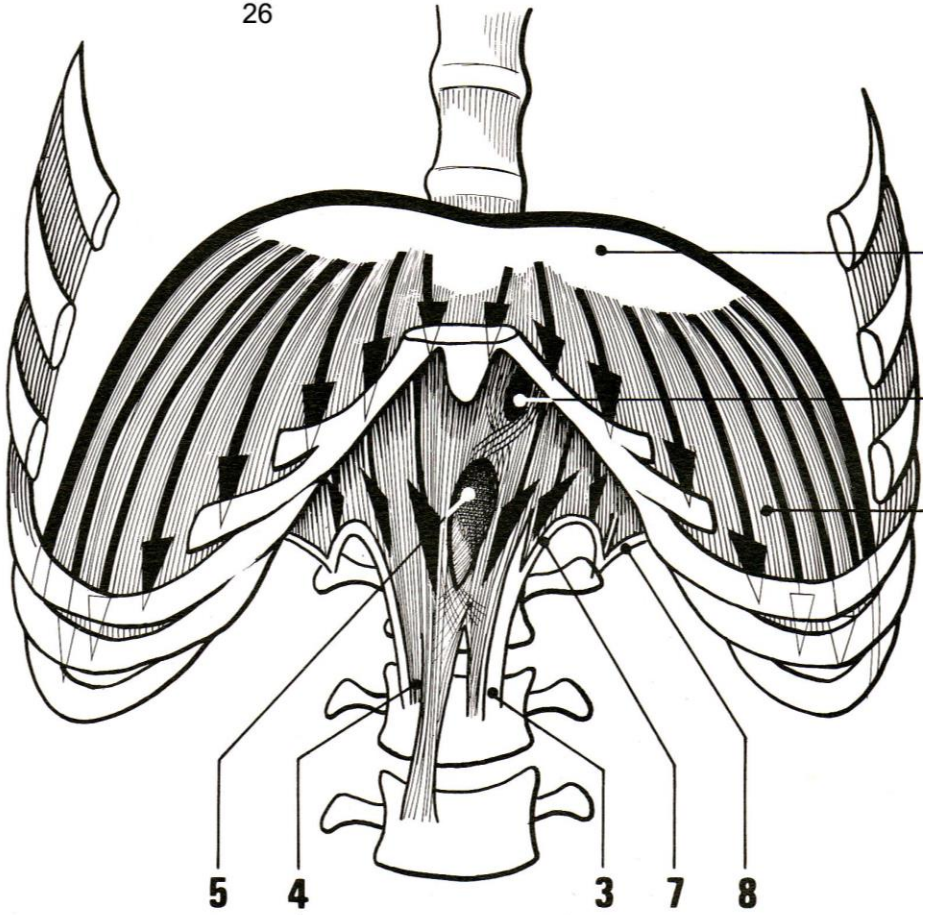
Overall movement of the thorax:

- The overall movement of the thorax is a complex process, requiring coordination of muscular and articular function.



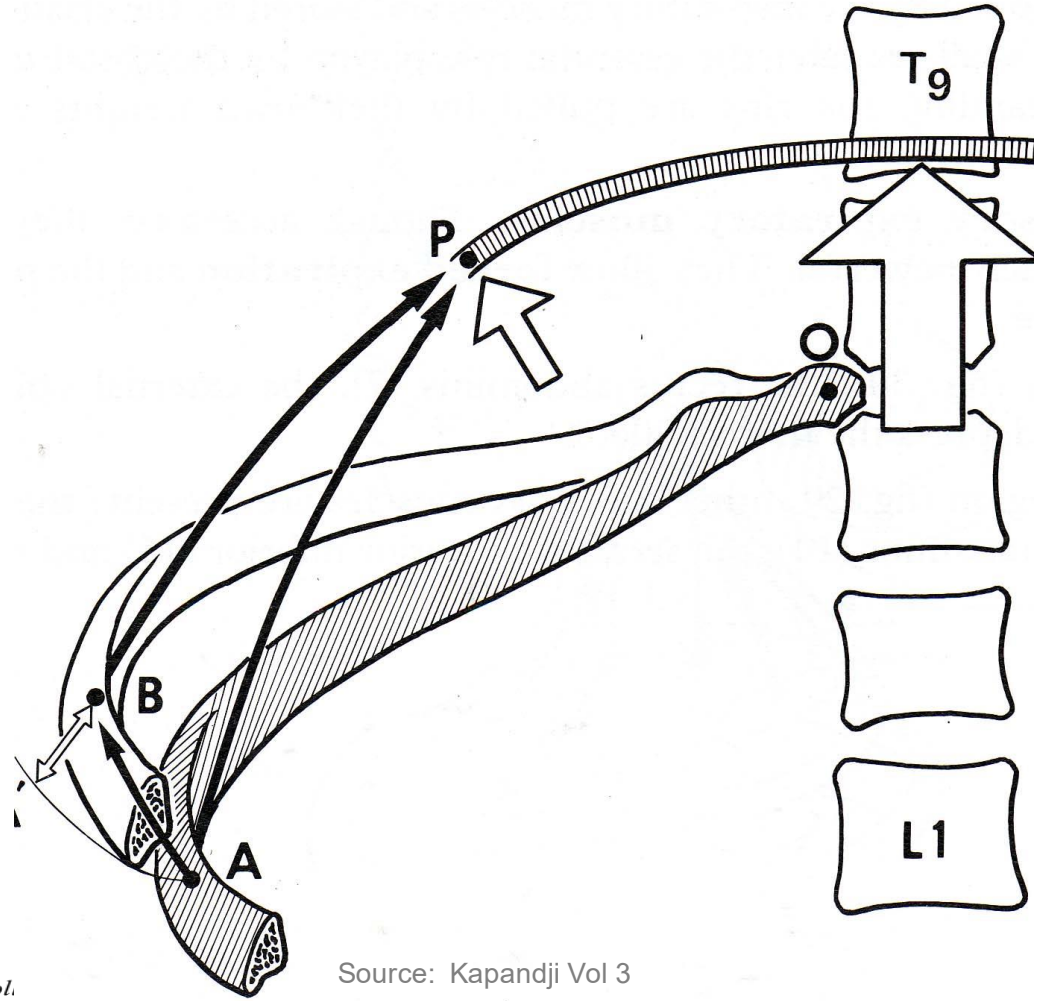
The diaphragm is a transverse septum which traverses the inside of the trunk. It must contract and produce expansion of the trunk. This requires a complex combination of forces and coordinated actions.

Source: Pernkopf Vol 2



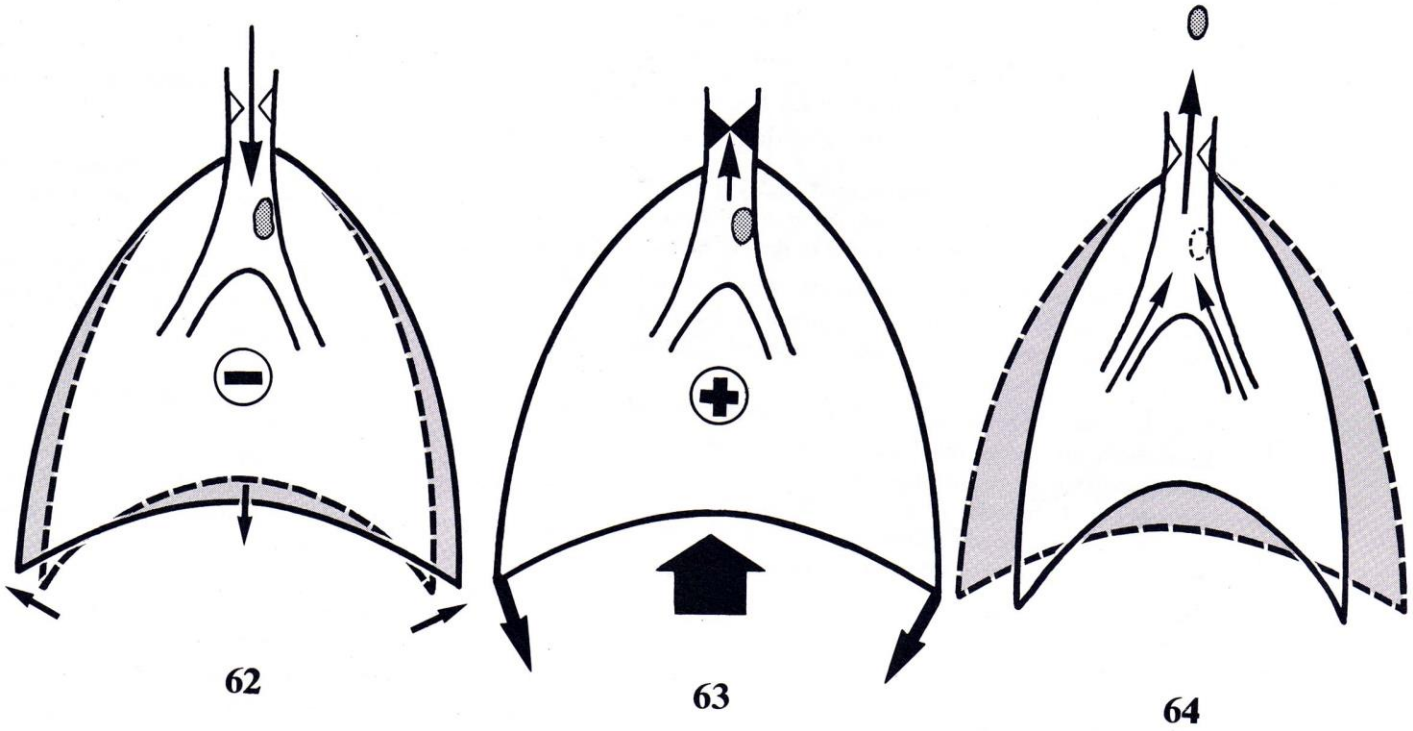
The vertical orientation of the fibers of the diaphragm helps produce the lateral expansion of the thorax with inhalation. This effect is severely limited when the diaphragm is flattened, as in asthma and COPD.

Source: Kapandji Vol 3



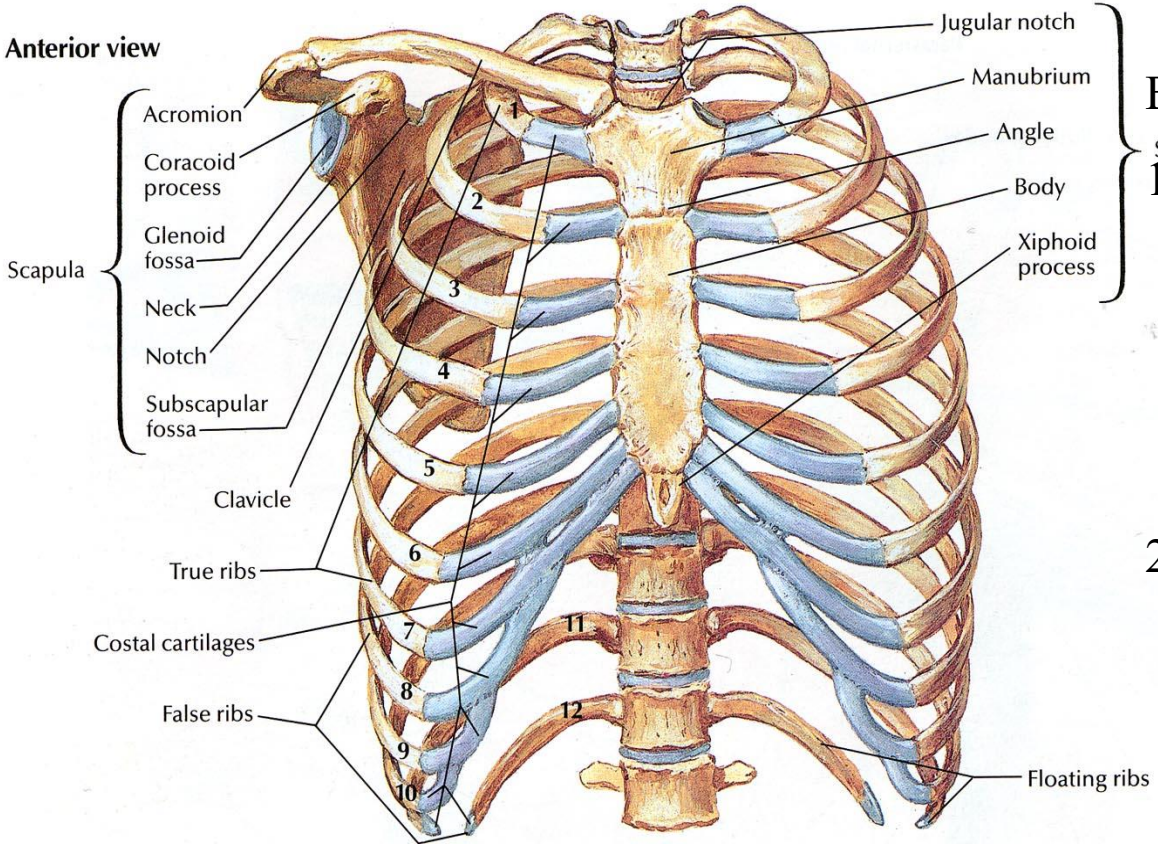
Source: Kapandji Vol 3

The articular relations of the rib and vertebrae cause the rib to swing in response to the pull of the diaphragm. This effect is enhanced by the simultaneous contraction of many accessory muscles of respiration. This will be affected by vertebral and rib dysfunction.



Flattening of the diaphragm will produce an inward movement of the thorax with inspiration (64, with dotted line as rest position). This inward movement of the rib cage is called paradoxical motion of the ribs.

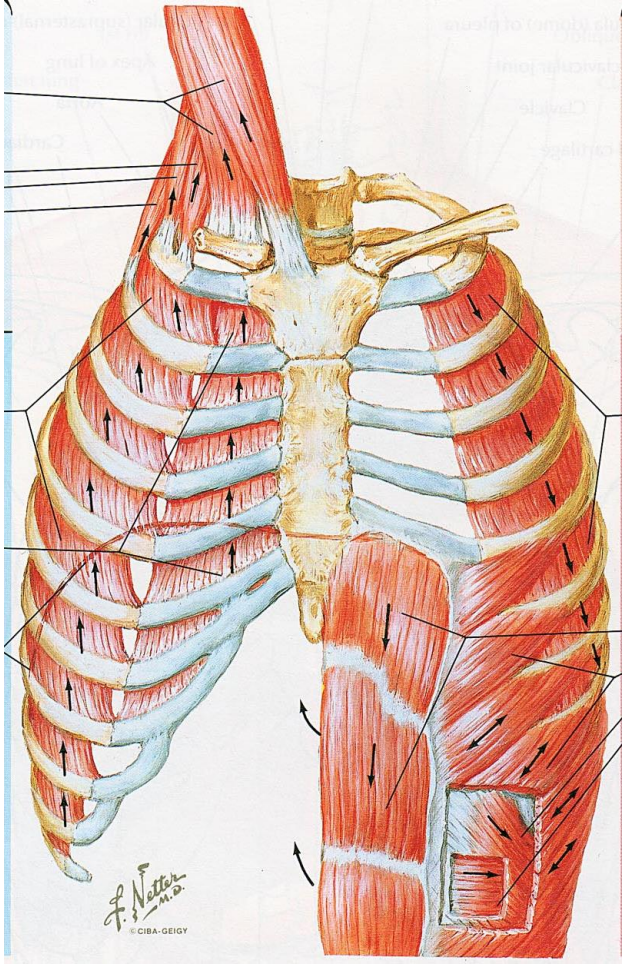
Source: Kapandji Vol 3



EVALUATION OF BREATHING

1. Thoracic cage excursion – observe motion
    1. Upper – midline – sternum, anterior ribs
    2. Lower – mid axillary
2. Thoracic cage compliance – motion testing

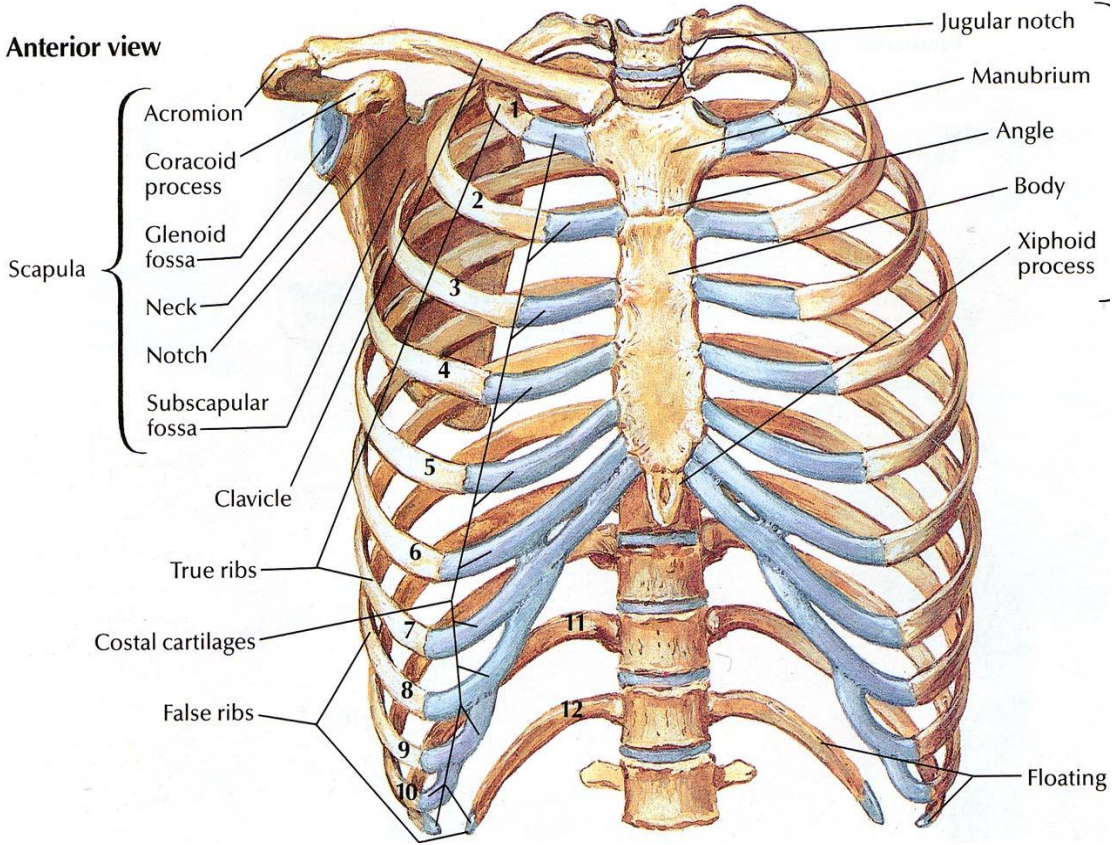
Source: Netter atlas of Human Anatomy



Source: Netter atlas of Human Anatomy

EVALUATION OF BREATHING

1. Respiratory muscles
    1. Fatigue
        1. Intercostal retractions
        2. Paradoxical motion
2. Hypertrophy/shortening
3. Resting tone



EVALUATION OF BREATHING

1. Overall pattern
2. Often visible when interviewing
3. Symmetry
    1. Inhalation/exhalation
    2. Upper/lower
    3. Side – side
    4. Effort

Source: Netter atlas of Human Anatomy

References

- Foundations of Osteopathic Medicine 4th Edition by Seffinger. Chapter 40 A. pg 923-944

Lecture Feedback Form: